Sinyal ve Sistemler

Sunu 1

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Signals and systems overview

What is Signal?

It is a representation of physical quantity (Sound, temperature, intensity, Pressure, etc..,) that varies concerning time or space or dependent or independent variable.

or

It is a single-valued function that carries information through Amplitude, Frequency, and Phase.

Examples: voice signals, video signals, signals on telephone wires, etc.

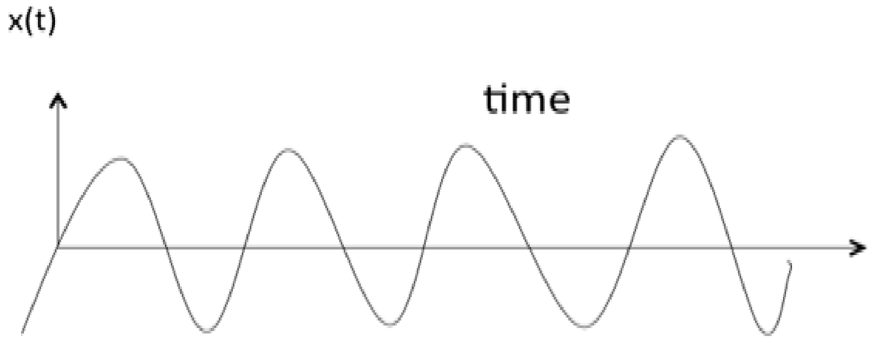
$$X(t) = A\,Sin(\omega t + \phi) = A\,Sin(2\pi f t + \phi)$$

Where A = Peak Amplitude

f = Frequency = $\frac{1}{T(Time\ Period)}$

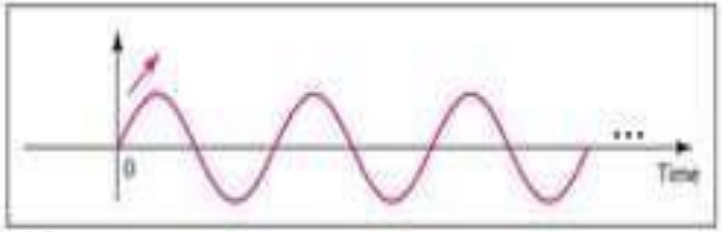
ϕ = Phase angle

Angular Frequency $\omega = 2\pi f$ (Linear Frequency)

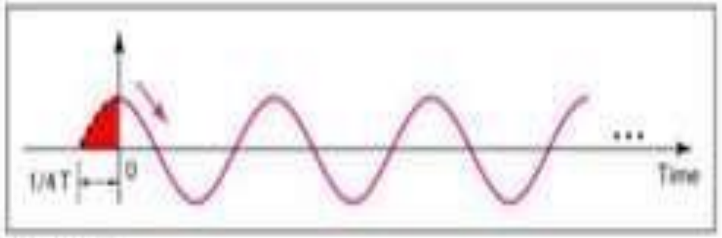


Signals and systems overview

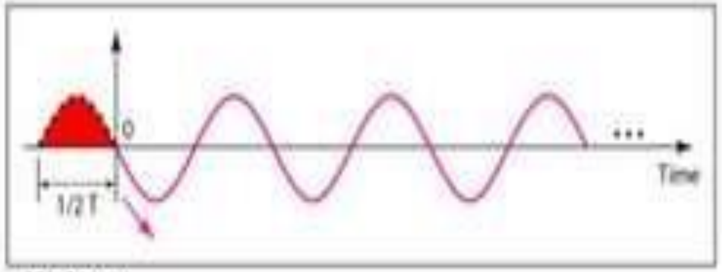
Signal with different Phases, Amplitudes and Frequencies



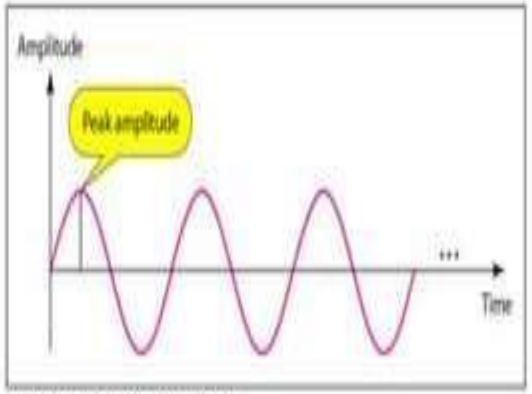
a. 0 degrees



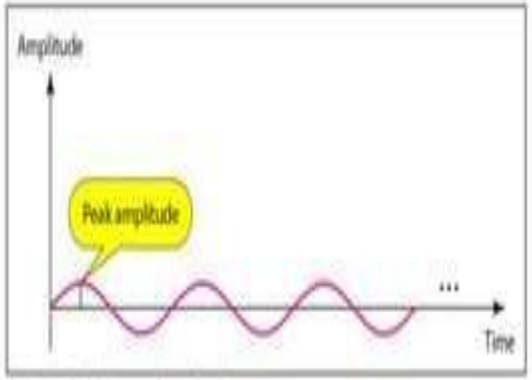
b. 90 degrees



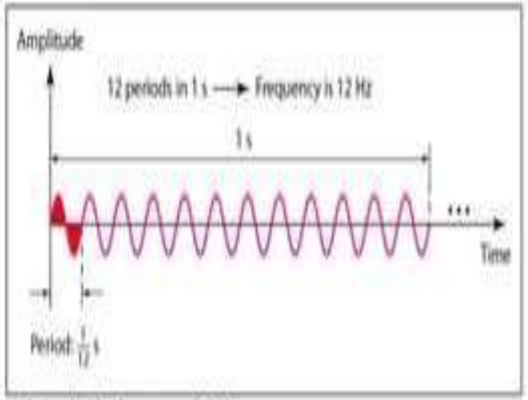
c. 180 degrees



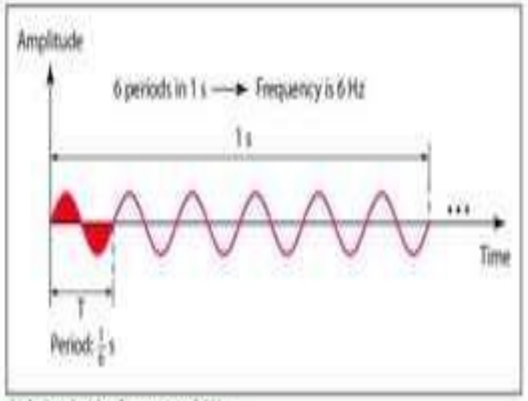
a. A signal with high peak amplitude



b. A signal with low peak amplitude



a. A signal with a frequency of 12 Hz



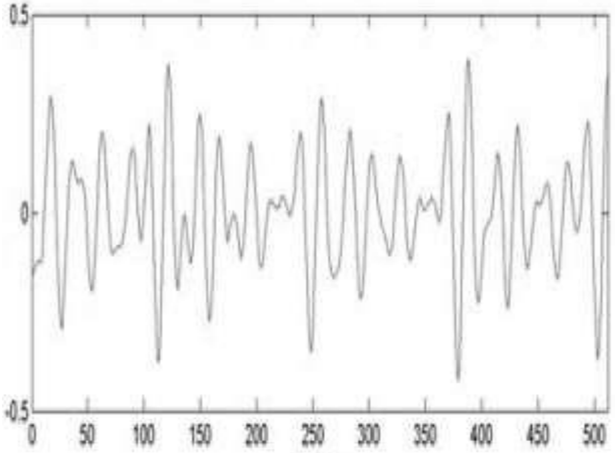
b. A signal with a frequency of 6 Hz

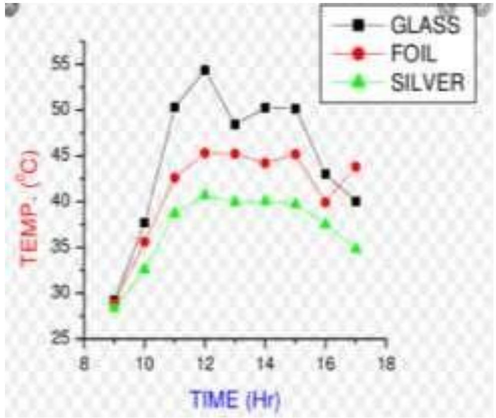
Classification of Signals

Types of Signals concerning the number of variables or dimensions

- **<u>One Dimensional or 1-D Signal:</u>** If the signal is a function of only one variable or If the Signal value varies for only one variable then it is called "One Dimensional or 1-D Signal"

Examples: Audio Signals, Biomedical Signals, temperature Signals, etc.., in which the signal is a function of "time"





Classification of Signals

- **Two Dimensional or 2-D Signal:** If the signal is a function of two variables or If the Signal value varies with two variables then it is called a “Two Dimensional or 2-D Signal”

Examples: Image Signal in which intensity is a function of two spatial co-ordinates “X” & “Y” i,.e I (X, Y)

- **Three Dimensional or 3-D Signal:** If the signal is a function of three variables or If the Signal value varies with three variables then it is called a “Three-dimensional or 3-D Signal”

Examples: Video Signal in which intensity is a function of two spatial co-ordinates “X” & “Y” and also time “t” i.e v(x,y,t)

Vector : 1D array

Image : 2D tensor
(for each color channel)

Video : 3D tensor
(for each color channel)

Classification of Signals

Types of Signal concerning the nature of the signal

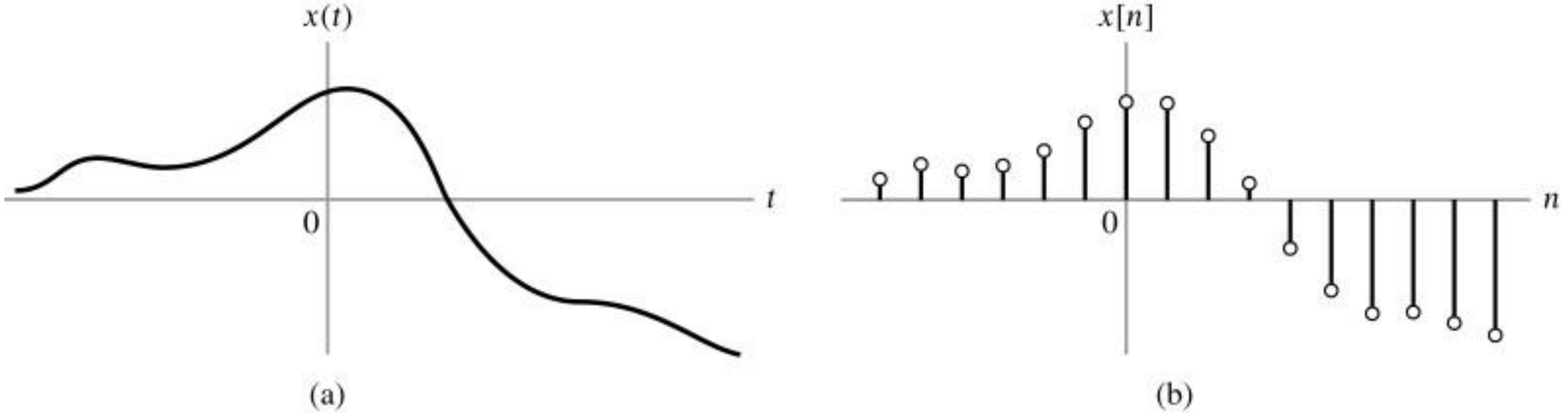
Continuous Time Signal (CTS) or Analog Signal :

If the signal values continuously varies with respect to time then it is called "Continuous Time Signal (CTS) or Analog Signal ". It contains infinite set of values and it is represented as shown below fig (a).

Digital Signal: If the signal contains only two values then it is called "Digital Signal". [0, 1]

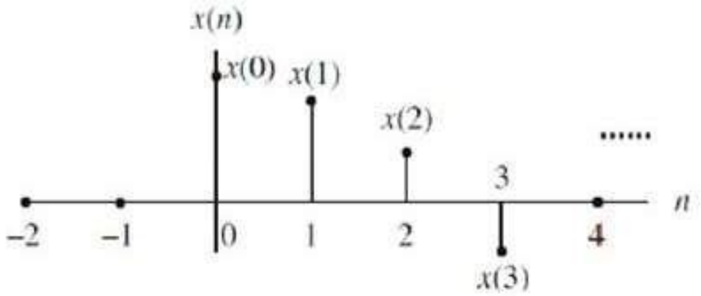
Discrete Time Signal (DTS):

If the signal contains a discrete set of values according to time then it is called "Discrete Time Signal (DTS)". It contains finite set of values. Sampling process converts Continuous time signal in to Discrete time signal.



Representation of Discrete Time Signal (DTS)

A discrete-time signal $x(n)$ is a function of an integer variable n. In the DS processor, the signal is represented by the discrete encoded values with a finite precision.



$$x(n) = \begin{cases} 1, & for\ n = 1,3 \\ 4, & for\ n = 2 \\ 0, & elsewhere \end{cases}$$

n	$\dots$	-2	-1	0	1	2	3	4	5	$\dots$
$x(n)$	$\dots$	0	0	0	1	4	1	0	0	$\dots$

Functional representation

Tabular representation

Graphical representation of a discrete-time signal $x(n)$

$x(n) = \{\dots, 0\ ,\mathbf{0}\ , 1\ , 4\ , 1, 0, 0, \dots\}$ $\quad$ $infinite - duration\ signal$

$x(n) = \{\mathbf{0}\ , -2\ , 1\ , 4\ , -1, \}$ $\quad$ $finite - duration\ signal$

Sequence representation (bold or arrow for origin n=0)

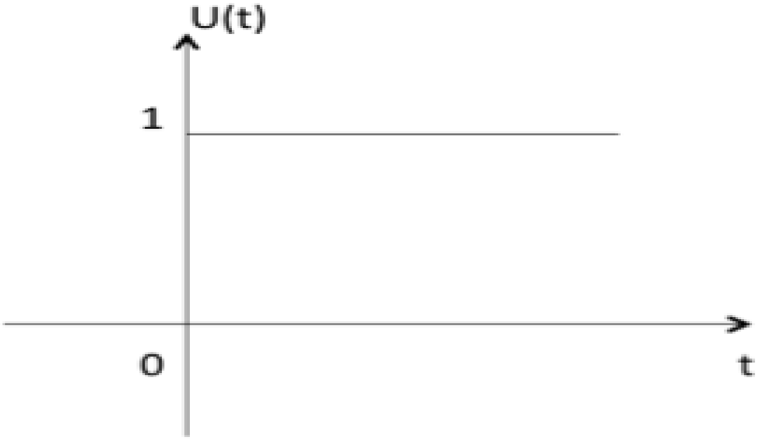
Mathematically a discrete-time signal $x(n)$ can be determined by

$$x(n) = x(t)|_{t=nT} = x(nT)$$

Basic Types of Signals

- **Unit Step Function**

Unit step function is denoted by u(t). It is defined as u(t) = 1 when $t \geq 0$
and 0 when $t < 0$



- It is used as best test signal.
- Area under unit step function is unity.

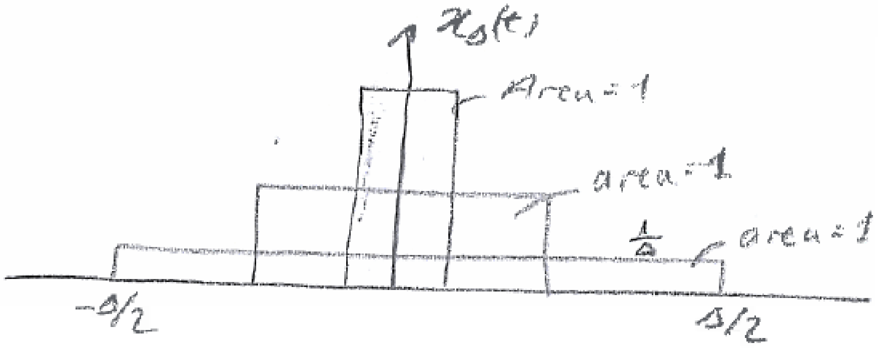
Basic Types of Signals

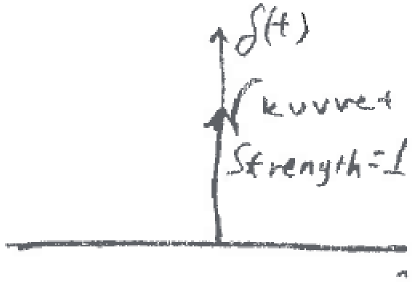
➢ Unit Impulse Function

Impulse function is denoted by δ(t). and it is defined as $\delta(t) = \begin{cases} 0; & t \neq 0 \\ 1; & t = 0 \end{cases}$ kuvvet

$$\delta(t) = 0 \quad \text{for } t \neq 0$$

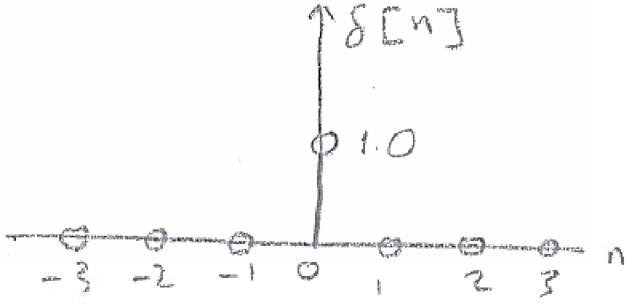
$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1$$





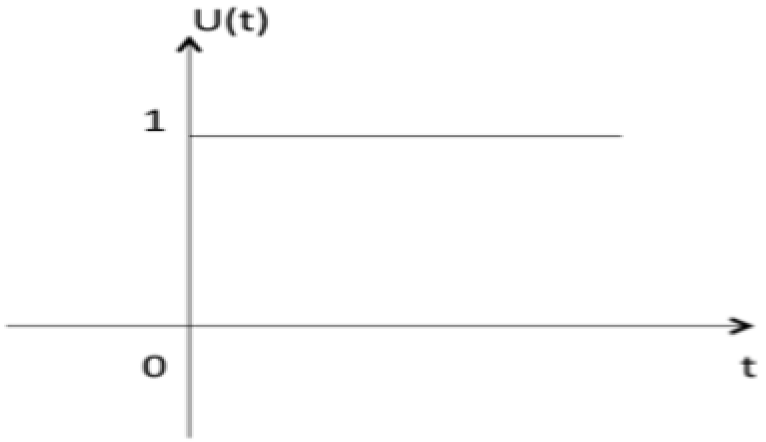
$$\lim_{\Delta \to 0} x_\Delta(t) = \delta(t)$$

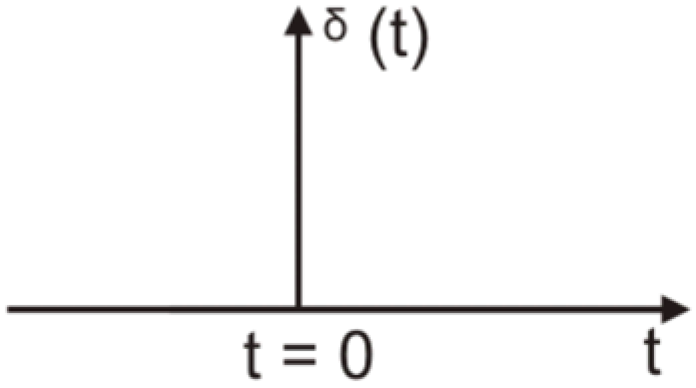
$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$



$$\delta(t) = \frac{d\,u(t)}{dt}$$

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau$$

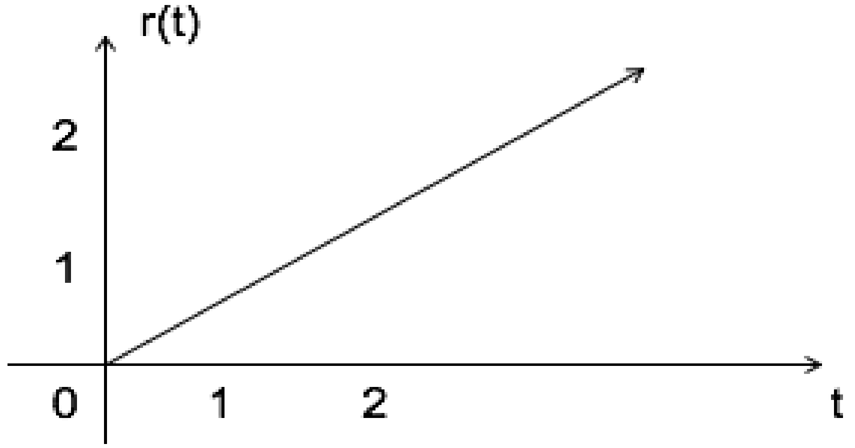




Basic Types of Signals

- **Ramp Signal**

Ramp signal is denoted by r(t), and it is defined as r(t) = $\begin{cases} t & t \geqslant 0 \\ 0 & t < 0 \end{cases}$



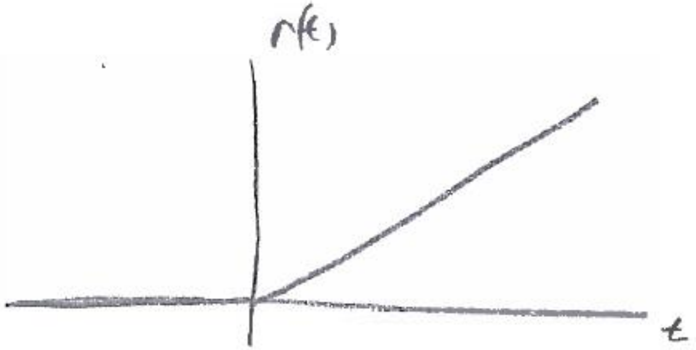
$$\int u(t) = \int 1 = t = r(t)$$

$$u(t) = \frac{dr(t)}{dt}$$

Area under unit ramp is unity.

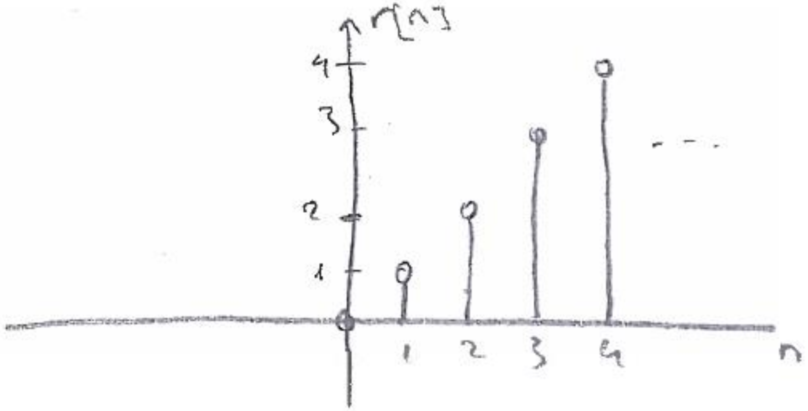
Ramp function (Rampa fonk.)

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0 & t < 0 \end{cases}$$



$$r(t) = t\, u(t)$$

$$r[n] = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

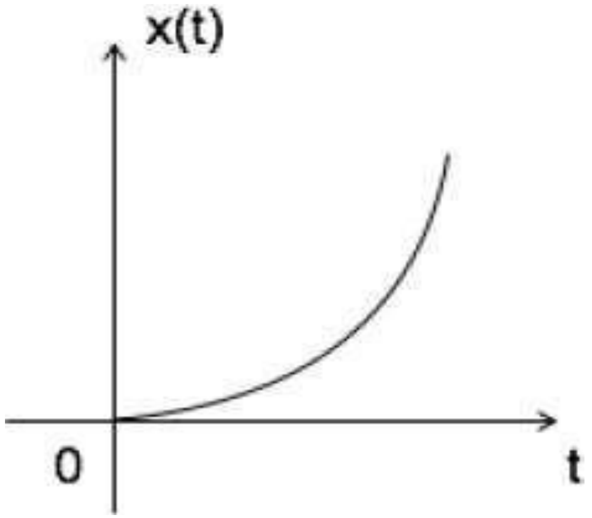


$$r[n] = n\, u[n]$$

Basic Types of Signals

- **Parabolic Signal**

Parabolic signal can be defined as x(t) = $\begin{cases} t^2/2 & t \geqslant 0 \\ 0 & t < 0 \end{cases}$



$$\iint u(t)dt = \int r(t)dt = \int tdt = \frac{t^2}{2} = parabolicsignal$$

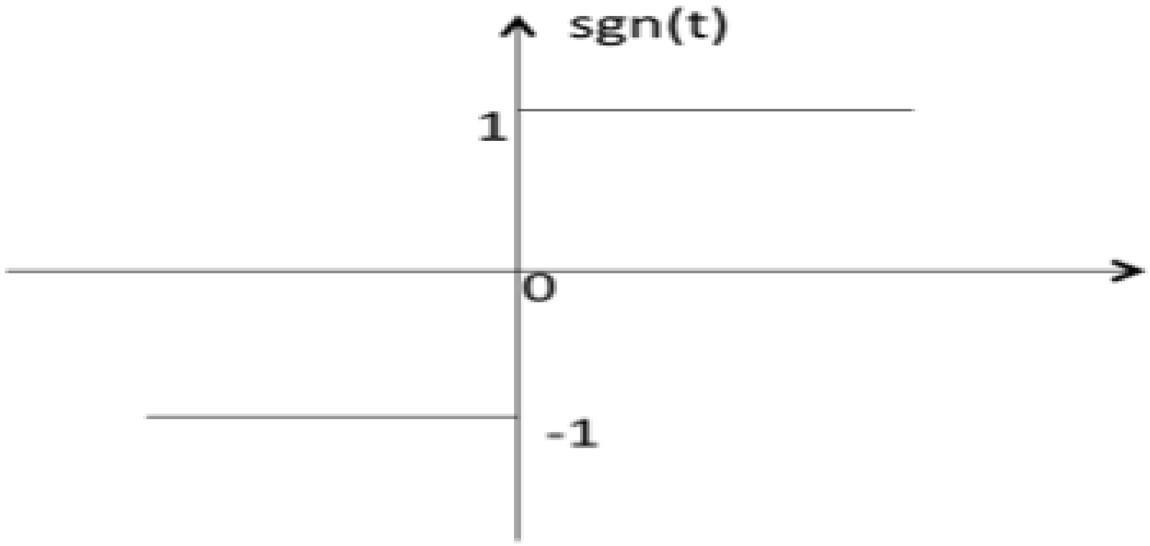
$$\Rightarrow u(t) = \frac{d^2x(t)}{dt^2}$$

$$\Rightarrow r(t) = \frac{dx(t)}{dt}$$

Basic Types of Signals

- **Signum Function**

Signum function is denoted as sgn(t). It is defined as sgn(t) = $\begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}$



Basic Types of Signals

- **Exponential Signal**

Exponential signal is in the form of $x(t) = e^{\alpha t}$

The shape of exponential can be defined by *α.*

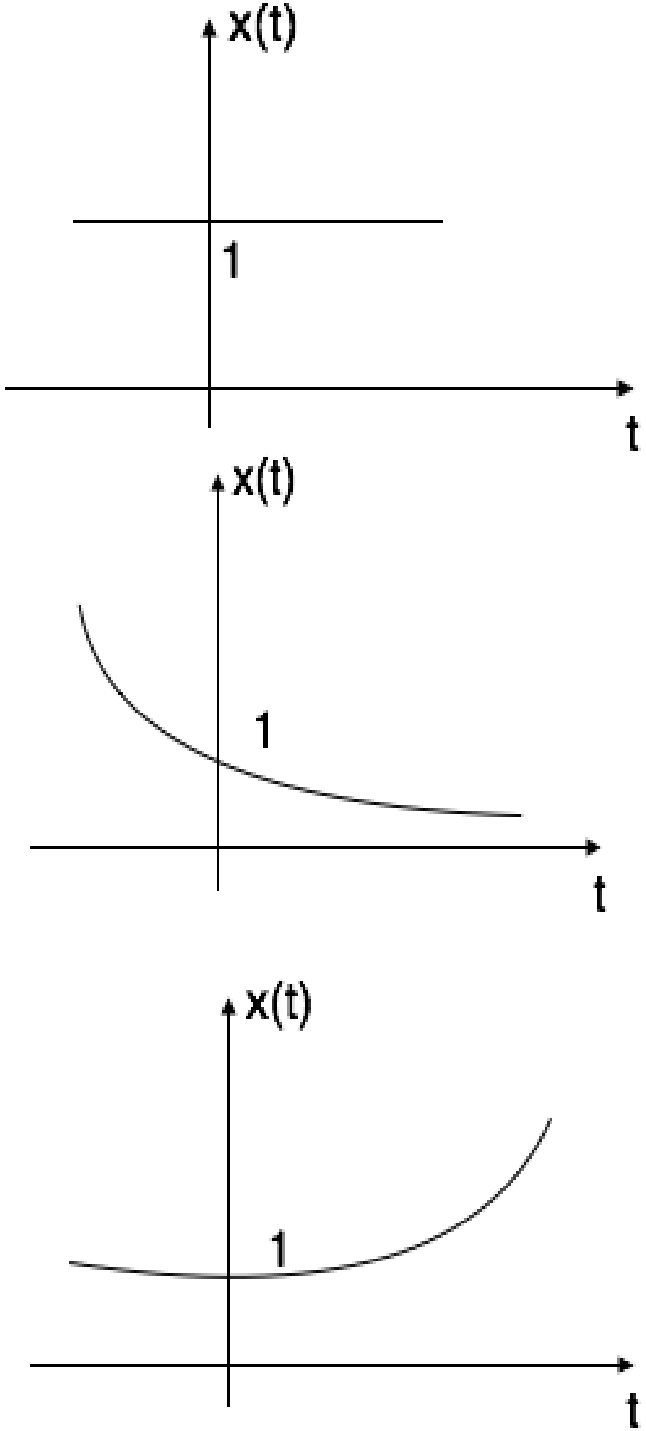
Case i: if $\alpha = 0 \rightarrow x(t) = e^0 = 1$

Case ii: if $\alpha < 0$ i.e. -ve then $x(t) = e^{-\alpha t}$,

The shape is called decaying exponential.

Case iii: if $\alpha > 0$ i.e. +ve then $x(t) = e^{\alpha t}$,

The shape is called raising exponential.



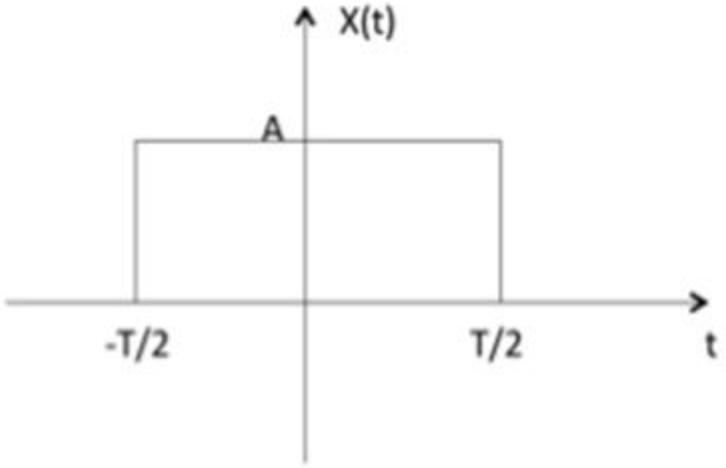
Basic Types of Signals

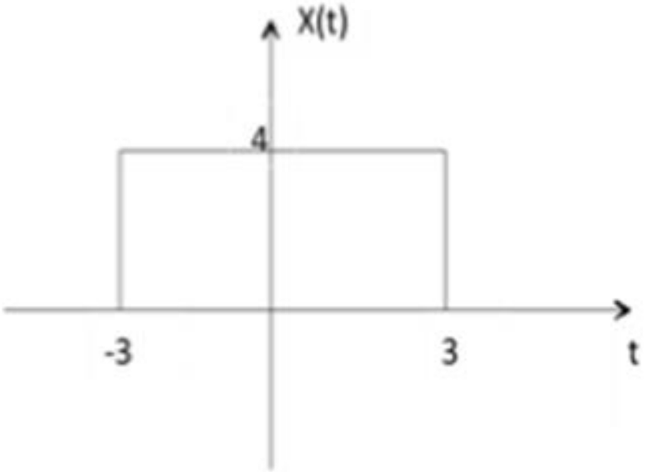
Rectangular Signal

Let it be denoted as x(t) and it is defined as

$$x(t) = A\,rect\left[\frac{t}{T}\right]$$

$$\text{ex: } 4\,rect\left[\frac{t}{6}\right]$$



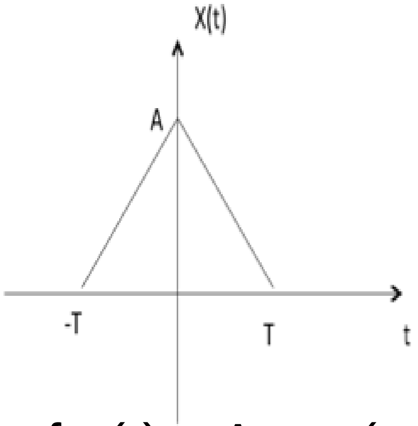


Basic Types of Signals

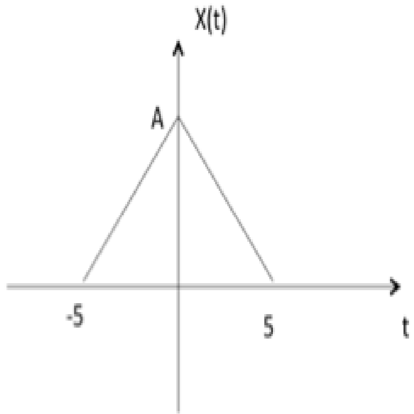
Triangular Signal

Let it be denoted as x(t),

$$x(t) = A\left[1 - \frac{|t|}{T}\right]$$

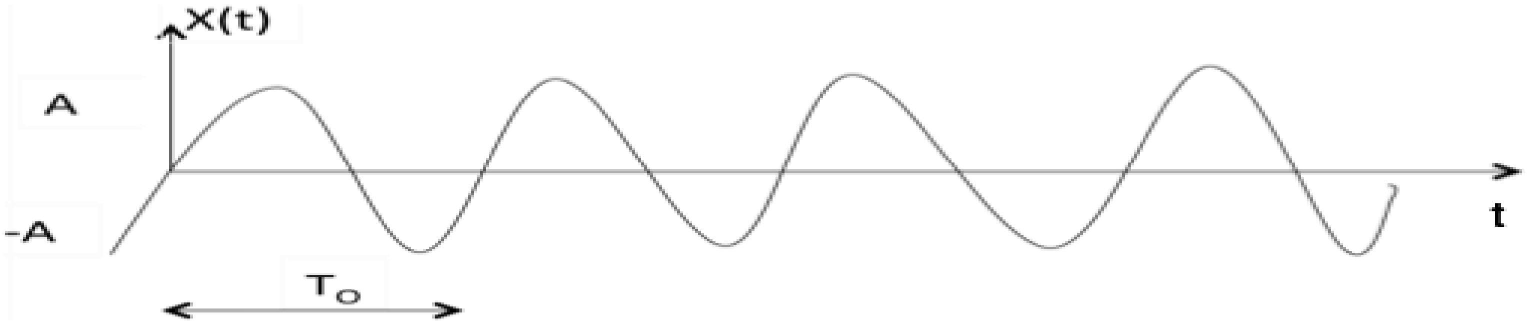


$$\text{ex:}\ x(t) = A\left[1 - \frac{|t|}{5}\right]$$



Sinusoidal Signal

Sinusoidal signal is in the form of x(t) = A cos($w_0 \pm \phi$) or A sin($w_0 \pm \phi$)



Where $T_0 = 2\pi/w_0$

Classification of Signals

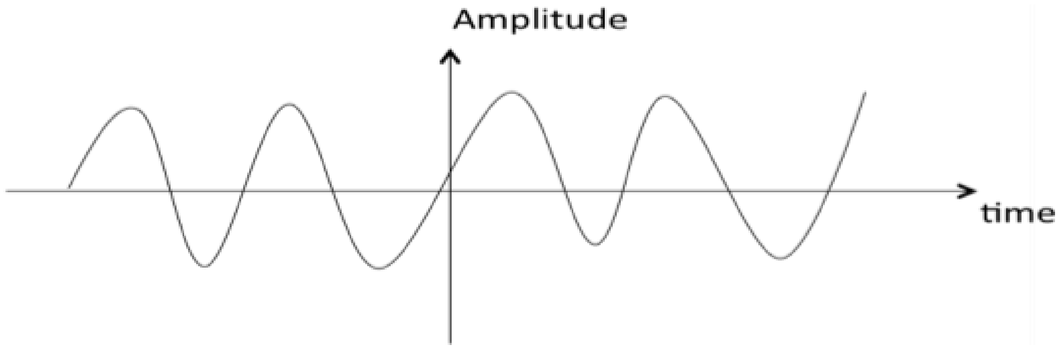
Signals are classified into the following categories:

- Continuous Time and Discrete Time Signals
- Deterministic and Non-deterministic Signals
- Even and Odd Signals
- Periodic and Aperiodic Signals
- Energy and Power Signals
- Real and Imaginary Signals

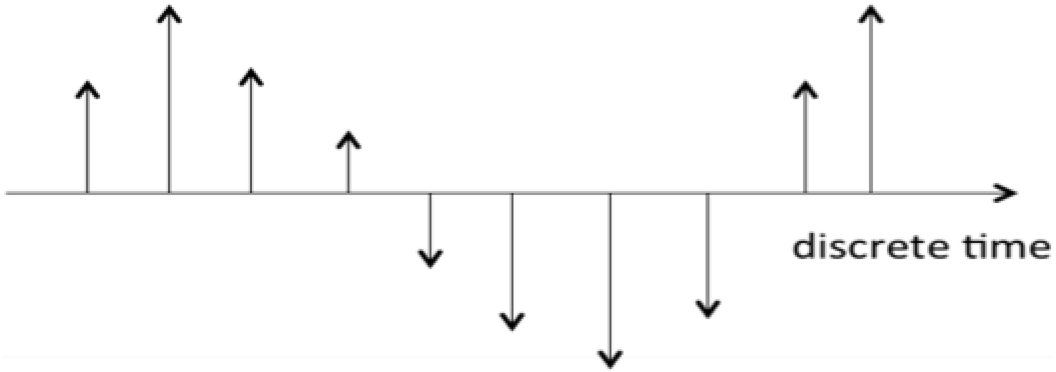
Classification of Signals

- **Continuous Time and Discrete Time Signals**

A signal is said to be continuous when it is defined for all instants of time.



A signal is said to be discrete when it is defined at only discrete instants of time.



- It is often useful to represent the signals graphically:

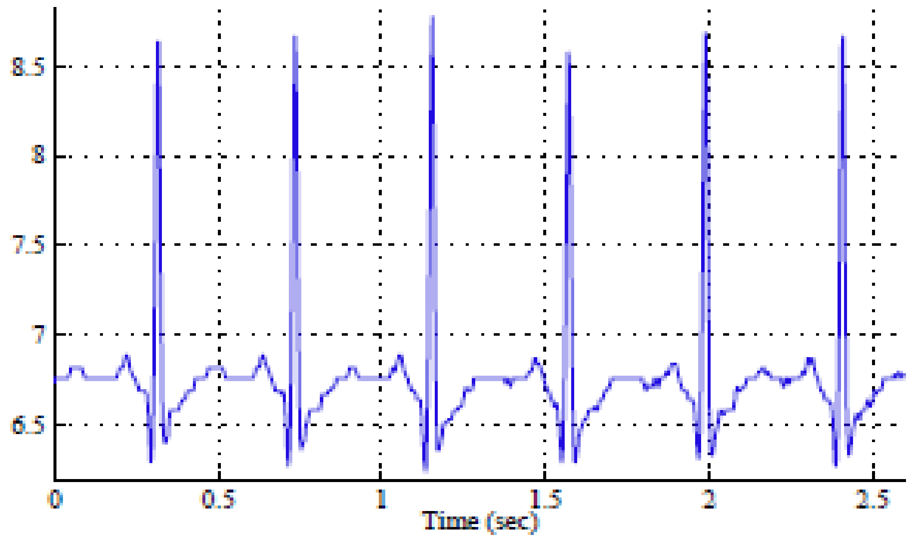


Figure 1: A continuous-time signal (electrocardiogram).

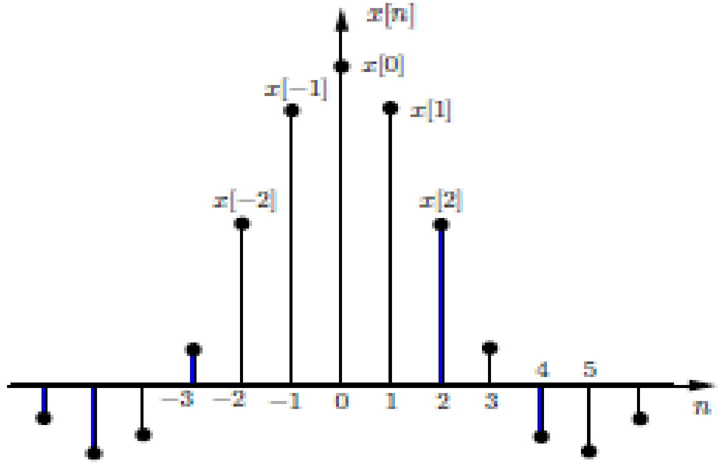


Figure 2: An example of discrete-time signal.

A discrete-time signal $x[n]$ can be defined in two ways:

1. We can specify a rule for calculating the nth value of the sequence. For example,

$$x[n] = x_n = \begin{cases} \left(\frac{1}{2}\right)^n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

or

$$\{x_n\} = \left\{1, \tfrac{1}{2}, \tfrac{1}{4}, \ldots, \left(\tfrac{1}{2}\right)^n, \ldots\right\}$$

2. We can also explicitly list the values of the sequence. For example, the sequence

$$\{x_n\} = \{\ldots, 0, 0, 1, 2, 2, \underset{\uparrow}{1}, 0, 1, 0, 2, 0, 0, \ldots\}$$

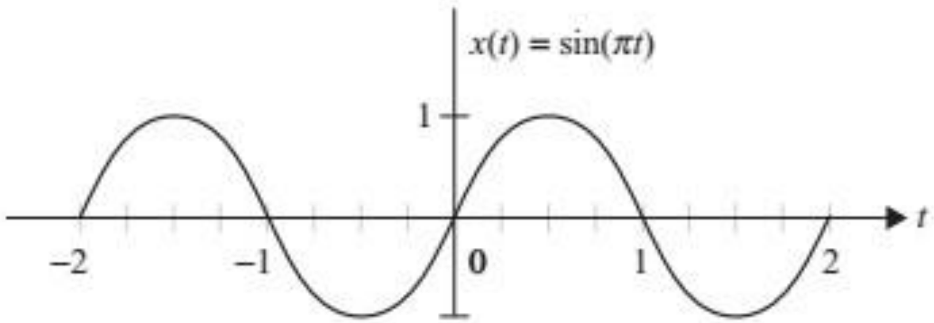
or

$$\{x_n\} = \{1, 2, 2, \underset{\uparrow}{1}, 0, 1, 0, 2\}$$

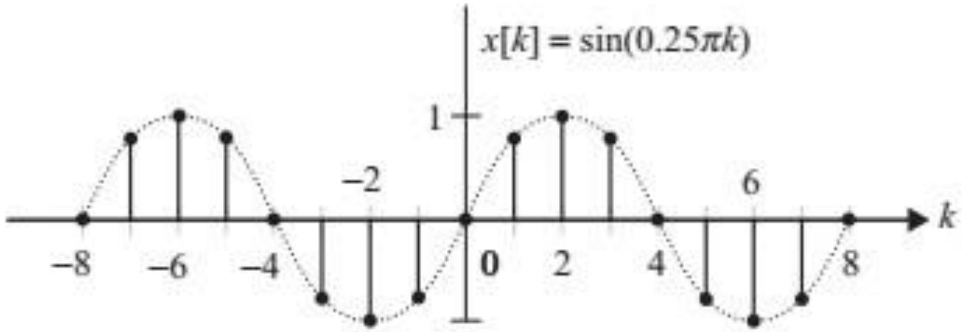
We use the arrow to denote the $n = 0$ term. We shall use the convention that if no arrow is indicated, then the first term corresponds to $n = 0$ and all the values of the sequence are zero for $n < 0$.

Example 1.1

Consider the CT signal $x(t) = \sin(\pi t)$ plotted in Fig. 1.3(a) as a function of time t. Discretize the signal using a sampling interval of $T = 0.25$ s, and sketch the waveform of the resulting DT sequence for the range $-8 \le k \le 8$.



(a)



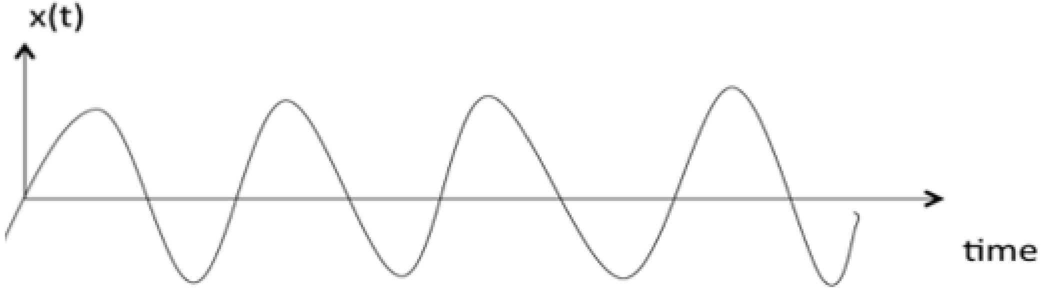
(b)

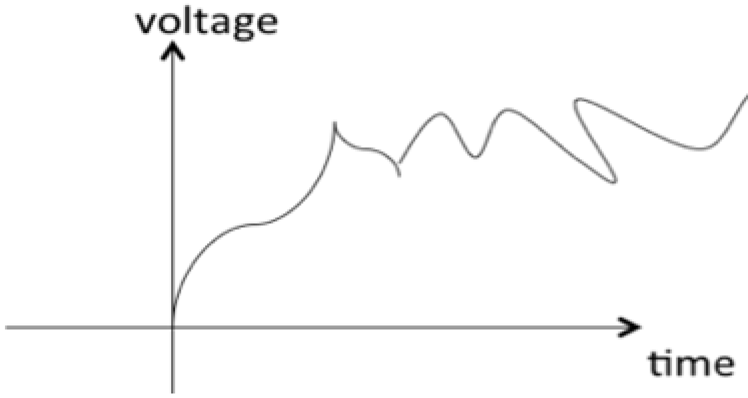
For $k = 0, \pm1, \pm2, \ldots$, the DT signal $x[k]$ has the following values:

$$x[-8] = x(-8T) = \sin(-2\pi) = 0, \qquad x[1] = x(T) = \sin(0.25\pi) = \frac{1}{\sqrt{2}},$$
$$x[-7] = x(-7T) = \sin(-1.75\pi) = \frac{1}{\sqrt{2}}, \qquad x[2] = x(2T) = \sin(0.5\pi) = 1,$$
$$x[-6] = x(-6T) = \sin(-1.5\pi) = 1, \qquad x[3] = x(3T) = \sin(0.75\pi) = \frac{1}{\sqrt{2}},$$
$$x[-5] = x(-5T) = \sin(-1.25\pi) = \frac{1}{\sqrt{2}}, \qquad x[4] = x(4T) = \sin(\pi) = 0,$$
$$x[-4] = x(-4T) = \sin(-\pi) = 0, \qquad x[5] = x(5T) = \sin(1.25\pi) = -\frac{1}{\sqrt{2}},$$
$$x[-3] = x(-3T) = \sin(-0.75\pi) = -\frac{1}{\sqrt{2}}, \qquad x[6] = x(6T) = \sin(1.5\pi) = -1,$$
$$x[-2] = x(-2T) = \sin(-0.5\pi) = -1, \qquad x[7] = x(7T) = \sin(1.75\pi) = -\frac{1}{\sqrt{2}},$$
$$x[-1] = x(-T) = \sin(-0.25\pi) = -\frac{1}{\sqrt{2}}, \qquad x[8] = x(8T) = \sin(2\pi) = 0,$$

Deterministic and Random Signals:

Deterministic signals are those signals whose values are completely specified for any given time. Thus, a deterministic signal can be modeled by a known function of time t. *Random* signals are those signals that take random values at any given time and must be characterized statistically. Random signals will not be discussed in this text.





Even and Odd signals

A signal is said to be **even** when it satisfies the condition $x(t) = x(-t)$

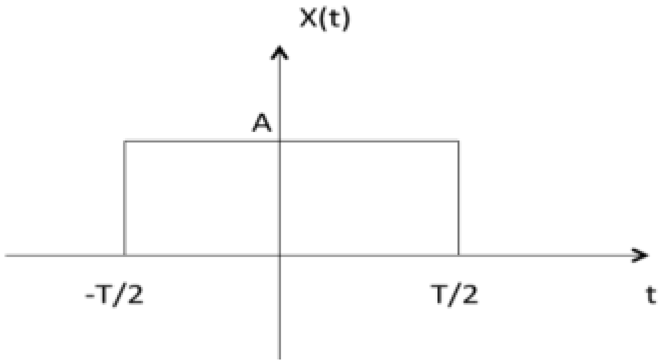
Example 1: t^2, t^4… cost etc.

Let $x(t) = t^2$

$x(-t) = (-t)^2 = t^2 = x(t)$

$\therefore t^2$ is even function

Example 2: As shown in the following diagram, rectangle function $x(t) = x(-t)$ so it is also even function.



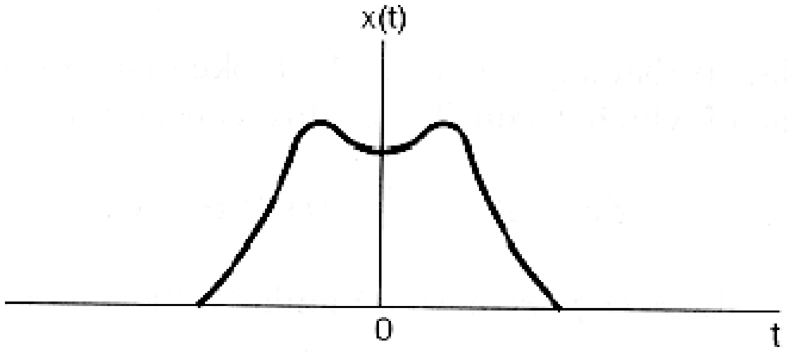
A signal is said to be **odd** when it satisfies the condition $x(t) = -x(-t)$

Even and Odd signals

Even signals:

$x(-t) = x(t)$, for continuous-time signals

$x[-n] = x[n]$, for discrete-time signals

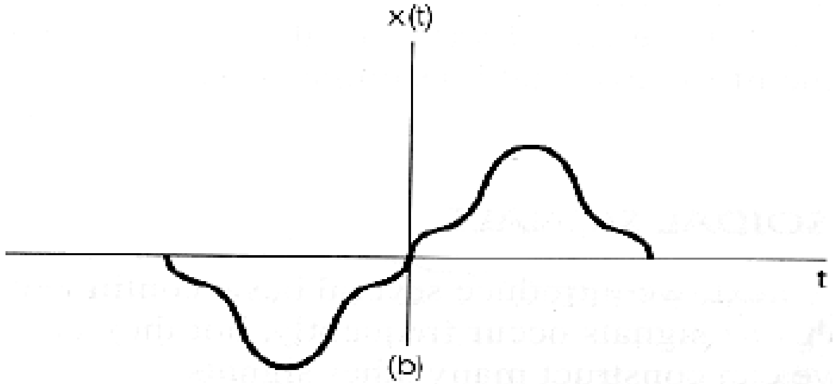


(a)

Odd signals:

$x(-t) = -x(t)$, for continuous-time signals

$x[-n] = -x[n]$, for discrete-time signals



(b)

Even and Odd signals

A CT signal $x_e(t)$ is said to be an even signal if

$$x_e(t) = x_e(-t).$$

Conversely, a CT signal $x_o(t)$ is said to be an odd signal if

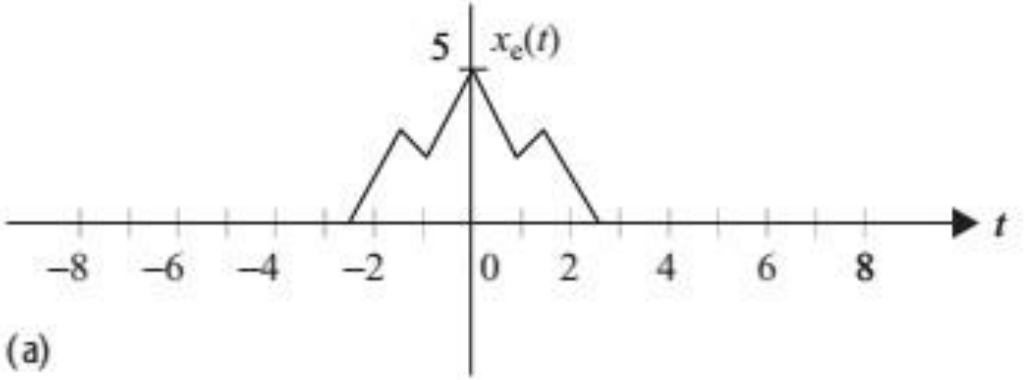
$$x_o(t) = -x_o(-t).$$

A DT signal $x_e[k]$ is said to be an even signal if

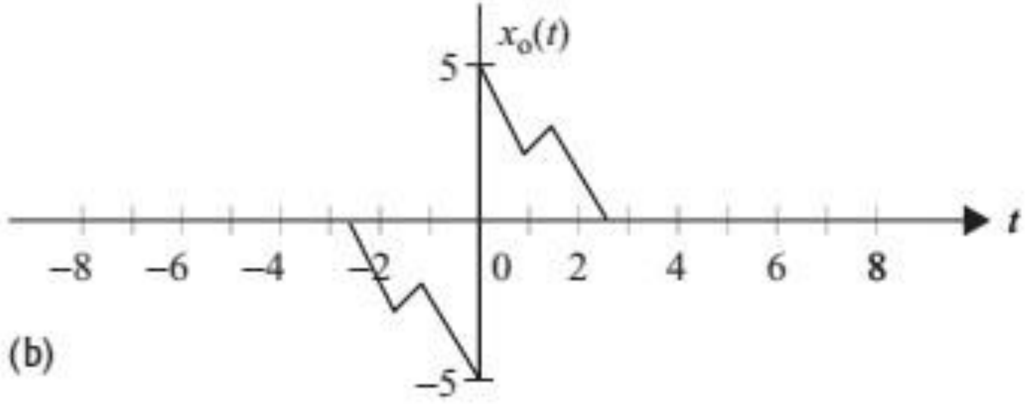
$$x_e[k] = x_e[-k].$$

Conversely, a DT signal $x_o[k]$ is said to be an odd signal if

$$x_o[k] = -x_o[-k].$$



(a)



(b)

Neither odd nor even signals can be expressed as a sum of even and odd signals as follows:

$$x(t) = x_e(t) + x_o(t),$$

where the even component $x_e(t)$ is given by

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)],$$

while the odd component $x_o(t)$ is given by

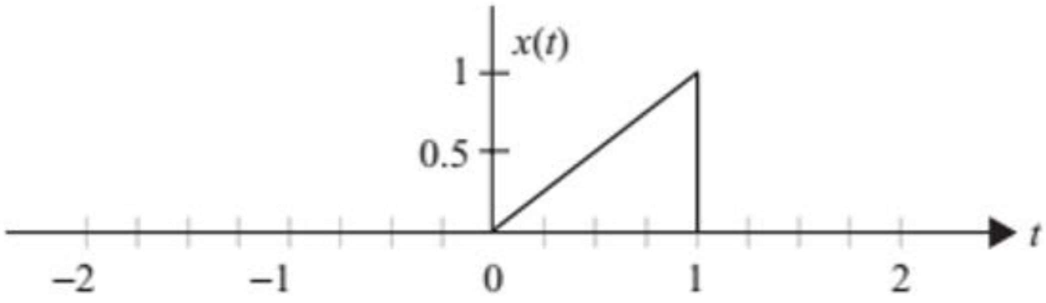
$$x_o(t) = \frac{1}{2}[x(t) - x(-t)].$$

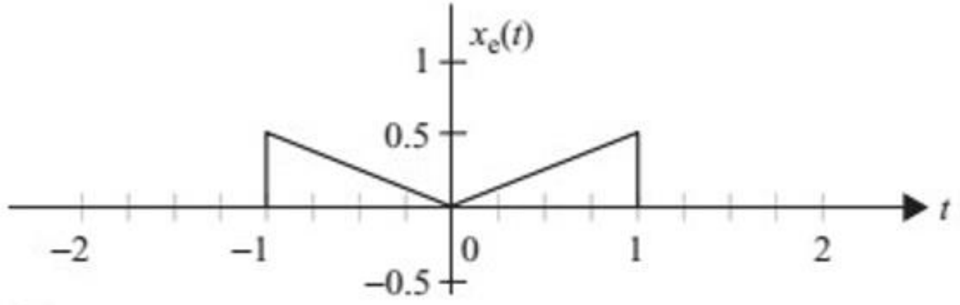
Example 1.9

Express the CT signal

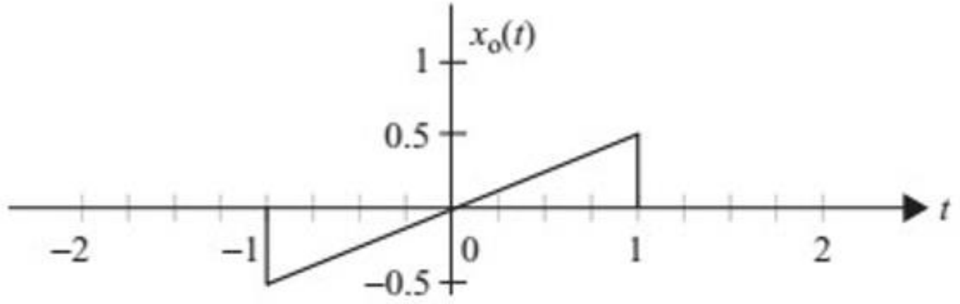
$$x(t) = \begin{cases} t & 0 \le t < 1 \\ 0 & \text{elsewhere} \end{cases}$$

as a combination of an even signal and an odd signal.





(b)



(c)

EXAMPLE 1.2 ANOTHER EXAMPLE OF EVEN AND ODD SIGNALS Find the even and odd components of the signal

$$x(t) = e^{-2t} \cos t.$$

Solution: Replacing t with $-t$ in the expression for $x(t)$ yields

$$x(-t) = e^{2t} \cos(-t)$$
$$= e^{2t} \cos t.$$

$$x_e(t) = \frac{1}{2}(e^{-2t} \cos t + e^{2t} \cos t) = \cosh(2t) \cos t$$

$$x_o(t) = \frac{1}{2}(e^{-2t} \cos t - e^{2t} \cos t) = -\sinh(2t) \cos t,$$

Exp: Find even and odd parts

$x(t) = 1 + t + 3t^2 + 5t^3 + 9t^4$ → Even: $1 + 3t^2 + 9t^4$ Odd: $t + 5t^3$

$x(t) = (1 + t^3) \cos^3(10t)$ → Even: $\cos^3(10t)$ Odd: $t^3 \cos^3(10t)$

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$$x(t) = (1 + t^3)\cos^3(10t) \quad \longrightarrow \quad \begin{array}{ll} \textbf{Even:} & \cos^3(10t) \\ \textbf{Odd:} & t^3\cos^3(10t) \end{array}$$

$$x(-t) = (1 - t^3)\cos^3(10(-t))$$

$$x_e(t) = \frac{1}{2}\left((1+t^3)\cos^3(10t) + (1-t^3)\cos^3(10t)\right)$$

$$= \frac{1}{2}\cos^3(10t)\,(1 + \cancel{t^3} + 1 - \cancel{t^3})$$

$$x_e(t) = \cos^3(10t)$$

$$x_o(t) = \frac{1}{2}(1+t^3)\cos^3(10t) - (1-t^3)\cos^3(10(-t))$$

$$= \frac{1}{2}\cos^3(10t)\,(\cancel{1} + t^3 - \cancel{1} + t^3)$$

$$= \frac{\cancel{2}}{\cancel{2}}\,t^3\cos^3(10t)$$

$$= t^3\cos^3(10t)$$

Real and Complex Signals:

A signal $x(t)$ is a *real* signal if its value is a real number, and a signal $x(t)$ is a *complex* signal if its value is a complex number. A general complex signal $x(t)$ is a function of the form

$$x(t) = x_1(t) + jx_2(t) \tag{1.1}$$

where $x_1(t)$ and $x_2(t)$ are real signals and $j = \sqrt{-1}$.

Note that in Eq. (*1.1*) t represents either a continuous or a discrete variable.

A signal is said to be **real** when it satisfies the condition x(t) = x*(t)

A signal is said to be **odd** when it satisfies the condition x(t) = -x*(t)

Example:

If x(t)= 3 then x*(t)=3*=3, here x(t) is a real signal.

If x(t)= 3j then x*(t)=3j* = -3j = -x(t), hence x(t) is a odd signal.

Note: For a real signal, imaginary part should be zero. Similarly for an imaginary signal, real part should be zero.

In the case of a complex-valued signal, we may speak of conjugate symmetry. A complex-valued signal $x(t)$ is said to be *conjugate symmetric* if

$$x(-t) = x^*(t), \tag{1.6}$$

where the asterisk denotes complex conjugation. Let

$$x(t) = a(t) + jb(t),$$

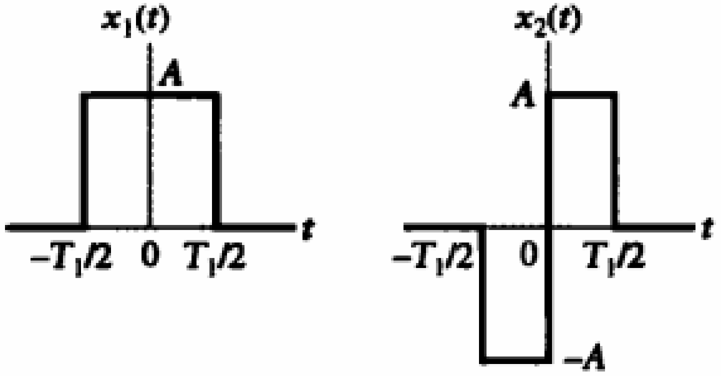
where $a(t)$ is the real part of $x(t)$, $b(t)$ is the imaginary part, and $j = \sqrt{-1}$. Then the complex conjugate of $x(t)$ is

$$x^*(t) = a(t) - jb(t).$$

Substituting $x(t)$ and $x^*(t)$ into Eq. (1.6) yields

$$a(-t) + jb(-t) = a(t) - jb(t).$$

▶ **Problem 1.2** The signals $x_1(t)$ and $x_2(t)$ shown in Figs. 1.13(a) and (b) constitute the real and imaginary parts, respectively, of a complex-valued signal $x(t)$. What form of symmetry does $x(t)$ have?



(a) **FIGURE 1.13** (b)

Complex-valued signal is conjugate symmetric

if its reel part even symmetric and its imaginary part is odd symmetric

The signal $x(t)$ is conjugate symmetric.

Periodic and aperiodic signals

A *periodic signal* $x(t)$ is a function of time that satisfies the condition

$$x(t) = x(t + T) \qquad \text{for all } t, \tag{1.7}$$

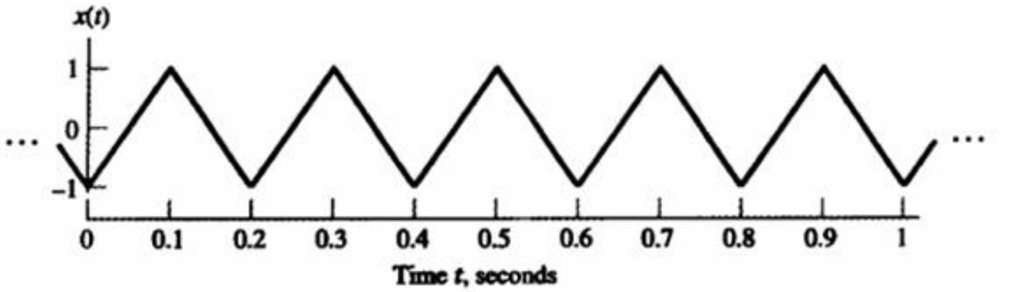
where T is a positive constant. Clearly, if this condition is satisfied for $T = T_0$, say, then it is also satisfied for $T = 2T_0, 3T_0, 4T_0, \ldots$. The smallest value of T that satisfies Eq. (1.7) is called the *fundamental period* of $x(t)$. Accordingly, the fundamental period T defines the duration of one complete cycle of $x(t)$. The reciprocal of the fundamental period T is called the *fundamental frequency* of the periodic signal $x(t)$; it describes how frequently the periodic signal $x(t)$ repeats itself. We thus formally write

$$f = \frac{1}{T}. \tag{1.8}$$

The frequency f is measured in hertz (Hz), or cycles per second. The *angular frequency*, measured in radians per second, is defined by

$$\omega = 2\pi f = \frac{2\pi}{T}, \tag{1.9}$$

► **Problem 1.3** Figure 1.15 shows a triangular wave. What is the fundamental frequency of this wave? Express the fundamental frequency in units of Hz and rad/s.



Answer: 5 Hz, or 10π rad/s.

The classification of signals into periodic and nonperiodic signals presented thus far applies to continuous-time signals. We next consider the case of discrete-time signals. A discrete-time signal $x[n]$ is said to be periodic if

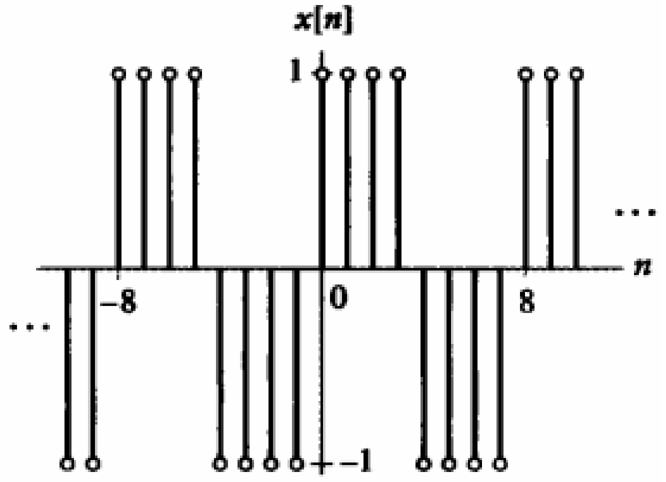
$$x[n] = x[n + N] \qquad \text{for integer } n, \tag{1.10}$$

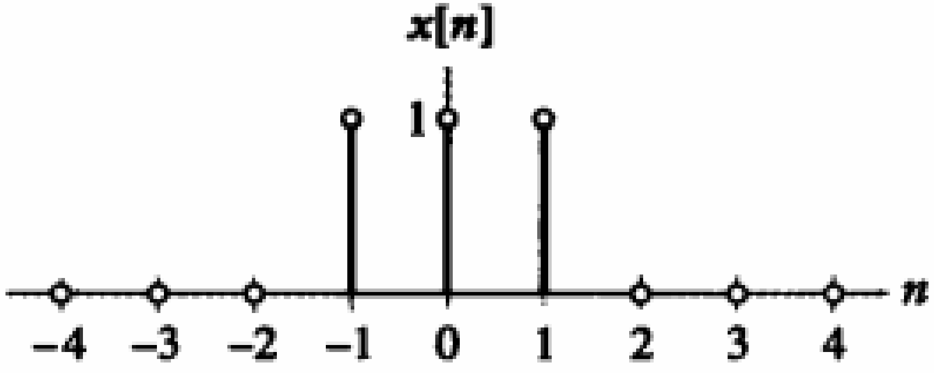
where N is a positive integer. The smallest integer N for which Eq. (1.10) is satisfied is called the fundamental period of the discrete-time signal $x[n]$. The fundamental angular frequency or, simply, fundamental frequency of $x[n]$ is defined by

$$\Omega = \frac{2\pi}{N}, \tag{1.11}$$

which is measured in radians.

The differences between the defining equations (1.7) and (1.10) should be carefully noted. Equation (1.7) applies to a periodic continuous-time signal whose fundamental period T has any positive value. Equation (1.10) applies to a periodic discrete-time signal whose fundamental period N can assume only a positive integer value.





Proposition 1.1 *An arbitrary DT sinusoidal sequence* $x[k] = A\sin(\Omega_0 k + \theta)$ *is periodic iff* $\Omega_0/2\pi$ *is a rational number.*

The term *rational number* used in Proposition 1.1 is defined as a fraction of two integers. Given that the DT sinusoidal sequence $x[k] = A\sin(\Omega_0 k + \theta)$ is periodic, its fundamental period is evaluated from the relationship

$$\frac{\Omega_0}{2\pi} = \frac{m}{K_0} \tag{1.7}$$

as

$$K_0 = \frac{2\pi}{\Omega_0} m. \tag{1.8}$$

Proposition 1.1 can be extended to include DT complex exponential signals. Collectively, we state the following.

(1) The fundamental period of a sinusoidal signal that satisfies Proposition 1.1 is calculated from Eq. (1.8) with m set to the smallest integer that results in an integer value for K_0.
(2) A complex exponential $x[k] = A\exp[\mathrm{j}(\Omega_0 k + \theta)]$ must also satisfy Proposition 1.1 to be periodic. The fundamental period of a complex exponential is also given by Eq. (1.8).

Example 1.4

Determine if the sinusoidal DT sequences (i)–(iv) are periodic:

(i) $f[k] = \sin(\pi k/12 + \pi/4)$;
(ii) $g[k] = \cos(3\pi k/10 + \theta)$;
(iii) $h[k] = \cos(0.5k + \phi)$;
(iv) $p[k] = e^{j(7\pi k/8+\theta)}$.

Solution

(i) The value of Ω_0 in $f[k]$ is $\pi/12$. Since $\Omega_0/2\pi = 1/24$ is a rational number, the DT sequence $f[k]$ is periodic. Using Eq. (1.8), the fundamental period of $f[k]$ is given by

$$K_0 = \frac{2\pi}{\Omega_0} m = 24m.$$

Setting $m = 1$ yields the fundamental period $K_0 = 24$.

To demonstrate that $f[k]$ is indeed a periodic signal, consider the following:

$$f[k + K_0] = \sin(\pi[k + K_0]/12 + \pi/4).$$

Substituting $K_0 = 24$ in the above equation, we obtain

$$\begin{aligned} f[k + K_0] &= \sin(\pi[k + K_0]/12 + \pi/4) = \sin(\pi k/12 + 2\pi + \pi/4) \\ &= \sin(\pi k/12 + \pi/4) = f[k]. \end{aligned}$$

(ii) $g[k] = \cos(3\pi k/10 + \theta)$;

(ii) The value of Ω_0 in $g[k]$ is $3\pi/10$. Since $\Omega_0/2\pi = 3/20$ is a rational number, the DT sequence $g[k]$ is periodic. Using Eq. (1.8), the fundamental period of $g[k]$ is given by

$$K_0 = \frac{2\pi}{\Omega_0}m = \frac{20m}{3}.$$

Setting $m = 3$ yields the fundamental period $K_0 = 20$.

(iii) $h[k] = \cos(0.5k + \phi)$;

(iii) The value of Ω_0 in $h[k]$ is 0.5. Since $\Omega_0/2\pi = 1/4\pi$ is not a rational number, the DT sequence $h[k]$ is not periodic.

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(iv) $p[k] = e^{j(7\pi k/8+\theta)}$

$$e^{i\pi} = \cos(\pi) + i\sin(\pi)$$

(iv) The value of Ω_0 in $p[k]$ is $7\pi/8$. Since $\Omega_0/2\pi = 7/16$ is a rational number, the DT sequence $p[k]$ is periodic. Using Eq. (1.8), the fundamental period of $p[k]$ is given by

$$K_0 = \frac{2\pi}{\Omega_0}m = \frac{16m}{7}.$$

Setting $m = 7$ yields the fundamental period $K_0 = 16$.

Energy and Power Signals

In electrical systems, a signal may represent a voltage or a current. Consider a voltage $v(t)$ developed across a resistor R, producing a current $i(t)$. The *instantaneous power* dissipated in this resistor is defined by

$$p(t) = \frac{v^2(t)}{R}, \tag{1.12}$$

or, equivalently,

$$p(t) = Ri^2(t). \tag{1.13}$$

In both cases, the instantaneous power $p(t)$ is proportional to the square of the amplitude of the signal. Furthermore, for a resistance R of 1 ohm, Eqs. (1.12) and (1.13) take on the same mathematical form. Accordingly, in signal analysis, it is customary to define power in terms of a 1-ohm resistor, so that, regardless of whether a given signal $x(t)$ represents a voltage or a current, we may express the instantaneous power of the signal as

$$p(t) = x^2(t). \tag{1.14}$$

On the basis of this convention, we define the *total energy* of the continuous-time signal $x(t)$ as

$$\begin{aligned} E &= \lim_{T\to\infty} \int_{-T/2}^{T/2} x^2(t)\,dt \\ &= \int_{-\infty}^{\infty} x^2(t)\,dt \end{aligned} \tag{1.15}$$

and its *time-averaged*, or *average*, *power* as

$$P = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)\, dt. \tag{1.16}$$

From Eq. (1.16), we readily see that the time-averaged power of a periodic signal $x(t)$ of fundamental period T is given by

$$P = \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)\, dt. \tag{1.17}$$

The square root of the average power P is called the *root mean-square* (rms) value of the periodic signal $x(t)$.

In the case of a discrete-time signal $x[n]$, the integrals in Eqs. (1.15) and (1.16) are replaced by corresponding sums. Thus, the total energy of $x[n]$ is defined by

$$E = \sum_{n=-\infty}^{\infty} x^2[n], \tag{1.18}$$

and its average power is defined by

$$P = \lim_{N \to \infty} \frac{1}{2N} \sum_{n=-N}^{N} x^2[n]. \tag{1.19}$$

Here again, from Eq. (1.19), the average power in a periodic signal $x[n]$ with fundamental period N is given by

$$P = \frac{1}{N}\sum_{n=0}^{N-1} x^2[n]. \tag{1.20}$$

A signal is referred to as an *energy signal* if and only if the total energy of the signal satisfies the condition

$$0 < E < \infty.$$

The signal is referred to as a *power signal* if and only if the average power of the signal satisfies the condition

$$0 < P < \infty.$$

▶ Problem 1.6

(a) What is the total energy of the rectangular pulse shown in Fig. 1.14(b)?

(b) What is the average power of the square wave shown in Fig. 1.14(a)?

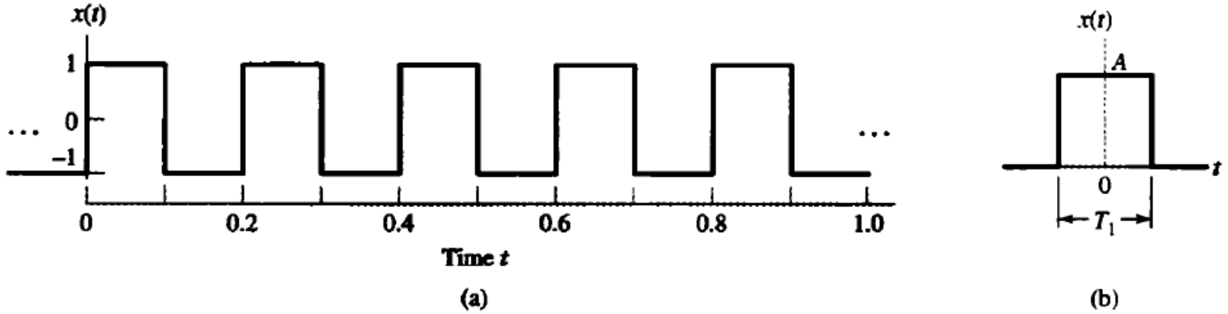
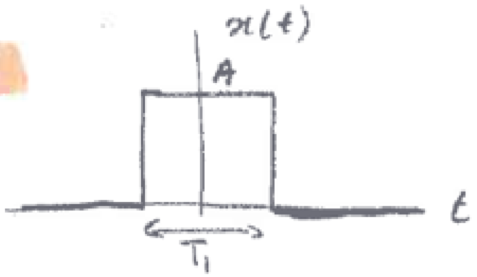


FIGURE 1.14 (a) Square wave with amplitude $A = 1$ and period $T = 0.2$ s. (b) Rectangular pulse of amplitude A and duration T_1.

Prob 1.6



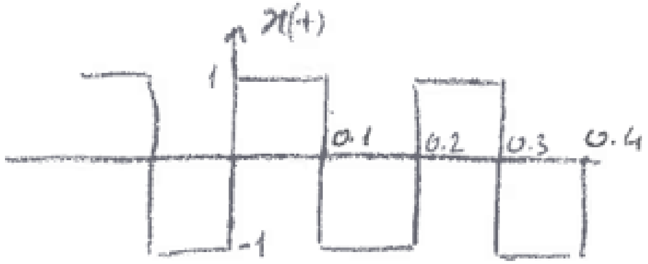
$$E = \int_{-\infty}^{\infty} x^2(t)\, dt$$

$$E = \int_{-\infty}^{+\infty} x^2(t)\, dt = \int_{-\frac{T_1}{2}}^{T_1/2} A^2\, dt = A^2 t \Big|_{-\frac{T_1}{2}}^{T_1/2}$$

$$= A^2\left(\frac{T_1}{2} - \left(-\frac{T_1}{2}\right)\right)$$

$$= AT_1$$

$$E = A^2 T_1$$



$$P = \frac{1}{T}\int_0^T x^2(t)\, dt$$

$$P = \frac{1}{T}\int_0^{0,2} x(t)^2\, dt$$

$$= \frac{1}{0,2}\left[\int_0^{0,1} (1)^2\, dt + \int_{0,1}^{0,2} (-1)^2\, dt\right]$$

$$= 1$$

▶ **Problem 1.8** Determine the total energy of the discrete-time signal shown in Fig. 1.17.

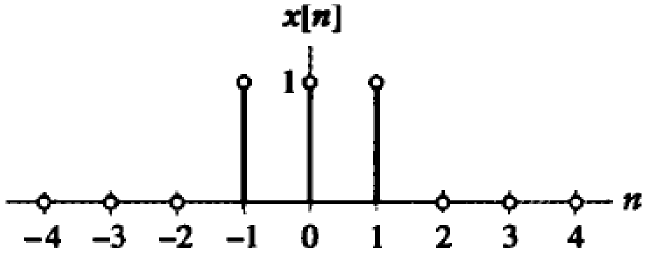


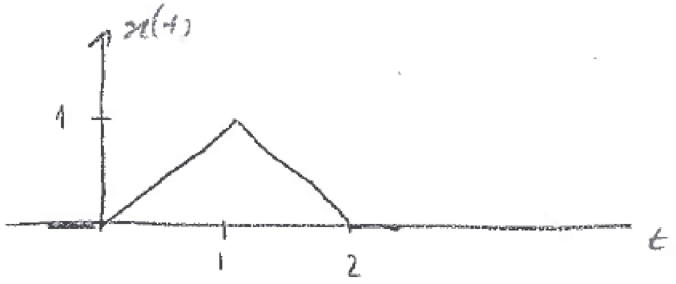
FIGURE 1.17 Nonperiodic discrete-time signal consisting of three nonzero samples.

$$E = \sum_{n=-\infty}^{\infty} x^2[n]$$

$$E = \sum_{n=-1}^{1} 1^2 = 3$$

▶ **Problem 1.9** Categorize each of the following signals as an energy signal or a power signal, and find the energy or time-averaged power of the signal:

(a) $x(t) = \begin{cases} t, & 0 \le t \le 1 \\ 2 - t, & 1 \le t \le 2 \\ 0, & \text{otherwise} \end{cases}$



$$E = \int_{-\infty}^{\infty} x^2(t)\, dt$$

$$E = \int_0^2 x^2(t)\, dt = \int_0^1 t^2\, dt + \int_1^2 (2-t)^2\, dt = \frac{t^3}{3}\Big|_0^1 + \int_1^2 4 - 4t + t^2\, dt$$

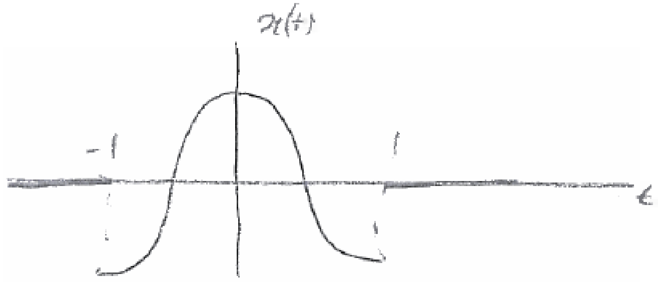
$$= \frac{1}{3} + \left(4t - \frac{4t^2}{2} + \frac{t^3}{3}\right)\Big|_1^2 = \cancel{\frac{1}{3}} + \left(\cancel{8} - \cancel{8} + \frac{8}{3}\right) - \left(4 - 2 + \frac{1}{3}\right)$$

$$= -2 + \frac{8}{3} = +\frac{2}{3}$$

Energy signal

▶ **Problem 1.9** Categorize each of the following signals as an energy signal or a power signal, and find the energy or time-averaged power of the signal:

(d) $x(t) = \begin{cases} 5\cos(\pi t), & -1 \le t \le 1 \\ 0, & \text{otherwise} \end{cases}$



$$E = \int_{-1}^{1} 25\cos^2 \pi t \, dt = 25 \int_{-1}^{1} \frac{1}{2}(1 + \cos 2\pi t)\, dt = 25\left[\frac{1}{2}t + \frac{1}{2\cdot 2\pi}\sin 2\pi t\right]\Bigg|_{-1}^{1} =$$

$$= 25\left(\left[\frac{1}{2} + \frac{1}{4\pi}\sin 2\pi\right] - \left[-\frac{1}{2} - \frac{1}{2\cdot 2\pi}\sin 2\pi\right]\right)$$

$$= 25\left(\frac{1}{2} + \frac{1}{2}\right)$$

$$= 25$$

Energy signal.